

Quartic Zero Repair: Semantic Head–Tail Addendum

Exact fuel, finite relation calculus, a depth barrier, and a regenerative Laurent normal form

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Abstract

We sharpen the coarse quartic zero-repair programme in four directions.

First, the correct finite-support convention gives boundary-phase rank five at parent degree two and rank six from degree three onward.

Second, every one-step question which reads only the first two coefficients of the children admits an exact support-independent head–tail quotient: all columns $u \geq 6$ collapse to one bulk symbol.

Third, the terminal-depth ambiguity can be removed exactly: t^r has zero-repair life precisely r . At the same time a support-independent coefficient obstruction proves that every raw hard-edge boundary state

$$G_0(A, C) = \frac{1}{2} + \frac{1}{2}AC$$

fails at depth four. Hence deep boundary closure at G_0 is impossible; the required phase correction must be anticipatory, before the raw boundary state is formed.

Finally, the critical Galton–Watson/Kovačević fixed point

$$g(t) = 1 - \sqrt{1 - t}$$

admits an exact finite Laurent relation calculus on the convolutional powers g^m . Thus all nonprimitive scales collapse to one finite algebraic grammar. The remaining obstruction is concentrated at the primitive $m = 1$ boundary. No $O(\log n)$ theorem is claimed in this addendum.

1 The coarse zero-repair relation

Throughout,

$$P_b = \frac{1}{64}(1, 12, 38, 12, 1), \quad Q_b = \frac{1}{64}(0, 16, 32, 16, 0),$$

so

$$P_b(t) - Q_b(t) = \frac{(1 - t)^4}{64}.$$

Let K_n be the set of probability PGFs admitting a depth- n zero-repair continuation. Thus K_0 is the set of all probability laws and

$$B \in K_{n+1}$$

iff there exist $B_x \in K_n$, $x = 0, \dots, 4$, such that

$$Q_b(t)B(t) = \sum_{x=0}^4 P_b(x)t^x B_x(t). \tag{ZR}$$

The sets K_n are convex and nested.

2 Correction to the finite-support phase rank

Suppose the parent has degree m . Since Q_b has effective support $\{1, 2, 3\}$, the transport columns are

$$0 \leq u \leq m + 3,$$

not $m + 4$.

Let M_m be the row/column marginal map on the admissible cells (x, u) with $u \geq x$, and let R_m be the nine-coordinate boundary readout

$$(H_1(B_0), H_3(B_0), H_1(B_2), H_3(B_2), H_1(B_4), H_3(B_4), D_{21}, D_{43}, D_{\text{sh}}).$$

Theorem 1 (Correct support-dependent phase rank – PROVED). *For the actual transport support,*

$$\dim R_m(\ker M_m) = \begin{cases} 2, & m = 0, \\ 4, & m = 1, \\ 5, & m = 2, \\ 6, & m \geq 3. \end{cases}$$

Proof. Three of the nine raw outputs are fixed by the marginals. For $m \geq 3$, six independent zero-marginal plaquette circulations are available:

$$C_{0,1}, \quad C_{1,2}, \quad C_{1,3}, \quad C_{1,4}, \quad C_{3,4}, \quad C_{3,5}.$$

On the phase coordinates

$$\Phi = (H_3(B_0), H_1(B_2), H_3(B_2), H_1(B_4), H_3(B_4), D_{\text{sh}})$$

their evaluation matrix is

$$\begin{pmatrix} 32 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{8}{19} & 0 & 0 & 0 & 0 \\ 0 & -\frac{8}{19} & -\frac{16}{19} & -\frac{8}{19} & 0 & 0 \\ 0 & 0 & 0 & 0 & -16 & 0 \\ 0 & 0 & 0 & 0 & -16 & -32 \\ 0 & 0 & -1 & 0 & 0 & 0 \end{pmatrix},$$

whose determinant is

$$\frac{1048576}{361} \neq 0.$$

Hence the rank is six.

For $m = 2$, column $u = 6$ does not exist, so $C_{3,5}$ disappears. Exact row reduction gives rank five. The cases $m = 0, 1$ give ranks two and four respectively. $\square$

3 Exact head–tail semantic contraction

Write

$$B_x(j) = \frac{\pi(x, x+j)}{P_b(x)}.$$

Consider a one-step specification which constrains only

$$B_x(0), \quad B_x(1),$$

possibly together with the exact terminal condition $B_x = t$.

Theorem 2 (Lossless head–tail quotient – PROVED). *For every such one-step specification, all transport columns*

$$u \geq 6$$

may be replaced by one bulk column $\star$ without changing feasibility.

Thus the infinite-support transport problem is exactly equivalent to a finite flow problem on

$$\boxed{\{0, 1, 2, 3, 4, 5, \star\}}.$$

Proof. For every $x \leq 4$ and $u \geq 6$,

$$u - x \geq 2.$$

Hence transport mass in these columns cannot affect either $B_x(0)$ or $B_x(1)$.

Moreover every $u \geq 6$ is causally accessible to every row. Consequently the tail part of the transport graph is complete bipartite. Any feasible original transport therefore projects to the $\star$ column.

Conversely, given a feasible quotient flow, split every $\star$ -row mass among the original columns $u \geq 6$ with their prescribed column totals. Since the tail graph is complete bipartite, this reconstruction always exists. Terminal rows simply have no edge to $\star$. $\square$

Remark 3. *This is the exact analogue of a semantic contraction: the large internal transport fiber is quotiented before any recursive reasoning is performed. No support cutoff is introduced.*

4 A five-branch LOW relation

For $\lambda \in [0, 1]$ put

$$\mathcal{C}_\lambda = \{B : B(0) = 0, [t]B \leq \lambda\}.$$

The previous CORE is

$$\mathcal{C} = \mathcal{C}_{13/16}.$$

Define the smaller head class

$$\boxed{\mathcal{L} := \mathcal{C}_{43/64}}.$$

Theorem 4 (LOW $\rightarrow$ five-branch CORE – EXACT FINITE). *If*

$$B \in \mathcal{L},$$

then there exists an exact five-branch zero-repair transport satisfying

$$\boxed{B_0, B_1, B_2, B_3 \in \mathcal{C}, \quad B_4 \in \mathcal{C} \cup \{t\}}.$$

Proof. Write $c = b_1$. Then

$$q_2 = \frac{c}{4}.$$

To put row 2 in CORE, column 2 cannot enter $(2, 2)$. Rows 0, 1 can absorb at most

$$P_b(0) + \frac{13}{16}P_b(1) = \frac{1}{64} + \frac{39}{256} = \frac{43}{256}.$$

Thus

$$q_2 \leq \frac{43}{256} \iff c \leq \frac{43}{64}.$$

After quotienting the tail to $\star$, the remaining statement is a finite capacitated Hall problem.

If $q_5 < 1/64$, use CORE rows on all five branches. The binding head cut is exactly column $\{2\}$ and has zero slack at $c = 43/64$; every other head cut has positive rational slack.

If $q_5 \geq 1/64$, put the entire row-4 mass at column 5, so $B_4 = t$, subtract $1/64$ from column 5, and apply the same finite Hall test to rows 0, 1, 2, 3. Again all cuts are nonnegative. Therefore the quotient flow exists, and the head–tail lifting theorem gives a full transport. $\square$

5 The MID obstruction is a relation, not a dead end

Let

$$\frac{43}{64} \leq c \leq \frac{3}{4}, \quad B(0) = 0, \quad [t]B = c.$$

Define

$$\eta(c) := \frac{16c - 1}{12}.$$

Theorem 5 (Exact MID escape relation – EXACT FINITE). *There exists a five-branch transport such that*

$$B_0, B_2, B_3 \in \mathcal{C}, \quad B_4 \in \mathcal{C} \cup \{t\},$$

while branch 1 remains hard-edge and satisfies

$$(B_1)_1 = \eta(c).$$

In particular

$$\frac{13}{16} \leq \eta(c) \leq \frac{11}{12}.$$

Proof. The coefficient-2 identity is

$$16c = (B_0)_2 + 12(B_1)_1 + 38(B_2)_0. \quad (\text{HB})$$

Keeping B_2 in CORE forces $(B_2)_0 = 0$. Maximizing the column-2 contribution of row 0 gives

$$(B_0)_2 = 1.$$

Hence

$$12(B_1)_1 = 16c - 1,$$

which is the stated value.

The remaining feasibility is again a finite head–tail max-flow problem. For $q_5 < 1/64$ take branch 4 in CORE; for $q_5 \geq 1/64$ take $B_4 = t$. All finite min-cuts are nonnegative throughout $c \in [43/64, 3/4]$. $\square$

Remark 6 (The central relation coordinate). *Equation (HB) should not be projected prematurely to a single state type:*

$$12(B_1)_1 + 38(B_2)_0 \geq 16c - 1.$$

It is the exact trade-off between linear mass exported to branch 1 and boundary mass exported to branch 2.

Thus the interval

$$43/64 < c < 3/4$$

does not say that the recursion fails. It says that the state-only CORE quotient has forgotten one relational degree of freedom.

6 Convolution is additive in causal depth

Theorem 7 (Convolution-depth addition – PROVED). *For all $m, n \geq 0$,*

$$A \in K_m, \quad C \in K_n \implies AC \in K_{m+n}.$$

Proof. Induct on $m + n$.

If $m > 0$, choose a one-step witness

$$Q_b A = \sum_x P_b(x) t^x A_x, \quad A_x \in K_{m-1}.$$

Multiplication by C gives

$$Q_b(AC) = \sum_x P_b(x) t^x (A_x C).$$

By induction

$$A_x C \in K_{m+n-1}.$$

Hence $AC \in K_{m+n}$.

The case $m = 0 < n$ is identical after splitting C . □

7 Terminal fuel is exact

For a finite PGF B let $\deg B$ denote its top occupied degree.

Lemma 8 (Degree loss on the all-4 history – PROVED). *If B has degree d , every branch-4 child satisfies*

$$\deg B_4 \leq d - 1.$$

Consequently

$$\boxed{B \in K_n \implies \deg B \geq n.}$$

Proof. The top degree of $Q_b B$ is at most $d + 3$. Since the branch-4 contribution is $t^4 B_4$,

$$4 + \deg B_4 \leq d + 3.$$

Iterate this inequality along the all-4 history. □

Theorem 9 (Exact terminal fuel – PROVED). *For integers $r, n \geq 0$,*

$$\boxed{t^r \in K_n \iff n \leq r.}$$

Proof. For $r \geq 1$ the exact interior split is

$$t^r \longrightarrow \left(t^{r+1}, t^r, \frac{3}{38} t^{r-1} + \frac{32}{38} t^r + \frac{3}{38} t^{r+1}, t^r, t^{r-1} \right).$$

Induction and convexity prove $t^r \in K_r$ and therefore $t^r \in K_n$ for $n \leq r$.

Conversely $\deg(t^r) = r$, so the degree lemma forbids membership in K_n for $n > r$. □

In particular,

$$\boxed{t \in K_1, \quad t \notin K_2.}$$

Thus the symbol t means exactly one remaining unit of causal fuel.

Together with convolution addition,

$$\boxed{B \in K_n \implies t^r B \in K_{n+r}.}$$

8 A universal boundary-atom depth barrier

Define

$$a_n := \sup\{B(0) : B \in K_n\}.$$

Theorem 10 (Boundary-atom recursion – PROVED). *For every $n \geq 1$,*

$$a_n \leq \frac{1}{16} + \frac{3}{4}a_{n-1}, \quad a_0 = 1.$$

Consequently

$$\boxed{a_n \leq \frac{1}{4} + \left(\frac{3}{4}\right)^{n+1}.$$

Proof. The coefficient of t^1 in (ZR) gives, for every feasible split,

$$16b_0 = (B_0)_1 + 12(B_1)_0.$$

If $B \in K_n$, then $B_1 \in K_{n-1}$, so

$$(B_1)_0 \leq a_{n-1},$$

while trivially $(B_0)_1 \leq 1$. Therefore

$$16b_0 \leq 1 + 12a_{n-1}.$$

Taking the supremum proves the recurrence. □

The first bounds are

$$a_1 \leq \frac{13}{16}, \quad a_2 \leq \frac{43}{64}, \quad a_3 \leq \frac{145}{256}, \quad a_4 \leq \frac{499}{1024}.$$

Since

$$\frac{499}{1024} < \frac{1}{2},$$

we obtain the key negative theorem.

Corollary 11 (Raw hard-edge boundary cannot be deep – PROVED). *If*

$$A(0) = C(0) = 0,$$

then

$$G_0(A, C) = \frac{1}{2} + \frac{1}{2}AC$$

satisfies

$$\boxed{G_0(A, C) \notin K_4.}$$

Remark 12. *Therefore the previously established three-step critical-channel peeling of G_0 cannot be promoted to a deep five-branch closure theorem.*

This is not a defect of the peeling construction. It is a theorem-level obstruction to the entire strategy “form raw G_0 , then repair it recursively”.

9 Interior grammar and the sticky deficit

Define

$$G_r(A, C) = \frac{1}{2}t^r + \frac{1}{2}AC.$$

Suppose

$$Q_b A = \sum_x P_b(x) t^x A_x.$$

Proposition 13 (Exact interior grammar – PROVED). *For every $r \geq 1$, $G_r(A, C)$ has the children*

$$\begin{aligned} &G_{r+1}(A_0, C), \\ &G_r(A_1, C), \\ &\frac{3}{38}G_{r-1}(A_2, C) + \frac{32}{38}G_r(A_2, C) + \frac{3}{38}G_{r+1}(A_2, C), \\ &G_r(A_3, C), \\ &G_{r-1}(A_4, C). \end{aligned}$$

Thus branches 1, 3 are not mysterious in the interior: they consume one factor level and preserve the buffer grade.

If N denotes the remaining product depth and

$$d := N - r,$$

then after one factor split

$$x = 0 : d \mapsto d - 2,$$

$$x = 1, 3 : d \mapsto d - 1,$$

$$x = 4 : d \mapsto d,$$

and the $x = 2$ mixture contains

$$d - 2, \quad d - 1, \quad d.$$

Hence

the interior deficit never increases.

The only sticky deficit-preserving channels are precisely the $x = 4$ channel and the $r - 1$ component inside $x = 2$.

10 Critical regenerative normal form

The previous results say that the missing operation must occur before the raw $r = 0$ boundary is created.

There is a natural exact normal form for this operation.

Let

$g(t) = 1 - \sqrt{1 - t}.$

Then

$$g = \frac{1}{2}t + \frac{1}{2}g^2,$$

or equivalently

$t = 2g - g^2.$

This is the critical binary Galton–Watson/Kovačević fixed point.

Theorem 14 (Finite Laurent grammar on g^m – PROVED). *For every integer $m \geq 1$, define*

$$\begin{aligned} C_0 &= tg^{m+1}, \\ C_1 &= \frac{7}{12}g^{m+1} + \frac{5}{12}tg^m, \\ C_2 &= \frac{4}{19}g^{m-1} + \frac{27}{76}g^{m+1} + \frac{27}{76}tg^{m-1} \\ &\quad + \frac{1}{19}tg^m + \frac{1}{38}tg^{m+1}, \\ C_3 &= g^m, \quad C_4 = g^{m-1}. \end{aligned}$$

Every C_x is a probability PGF and

$$\boxed{Q_b(t)g(t)^m = \sum_{x=0}^4 P_b(x)t^x C_x(t).} \quad (\text{LG})$$

Proof. All displayed coefficients are nonnegative and sum to one in every child.

After dividing (LG) by g^m and substituting

$$t = 2g - g^2,$$

both sides become finite Laurent polynomials in g . Direct collection of powers gives the identity exactly.

For example the branch-2 multiplier is

$$\frac{4}{19}g^{-1} + \frac{27}{76}g + \frac{27}{76}tg^{-1} + \frac{1}{19}t + \frac{1}{38}tg.$$

Its coefficients sum to

$$\frac{16 + 27 + 27 + 4 + 2}{76} = 1.$$

□

Remark 15 (Semantic meaning). *The entire infinite family of nonprimitive scales g^m has collapsed to a single finite relation alphabet:*

$$\boxed{m \mapsto m-1, \ m, \ m+1}$$

together with at most one deterministic t -fuel factor.

Thus the scale problem is not an infinite collection of unrelated transport LPs. It has an exact finite Laurent calculus.

Remark 16 (The primitive obstruction). *The relation is not yet an infinite finite-mean construction. Along the all-4 history*

$$g^m \rightarrow g^{m-1} \rightarrow \cdots \rightarrow g^0,$$

and

$$g^0 = 1$$

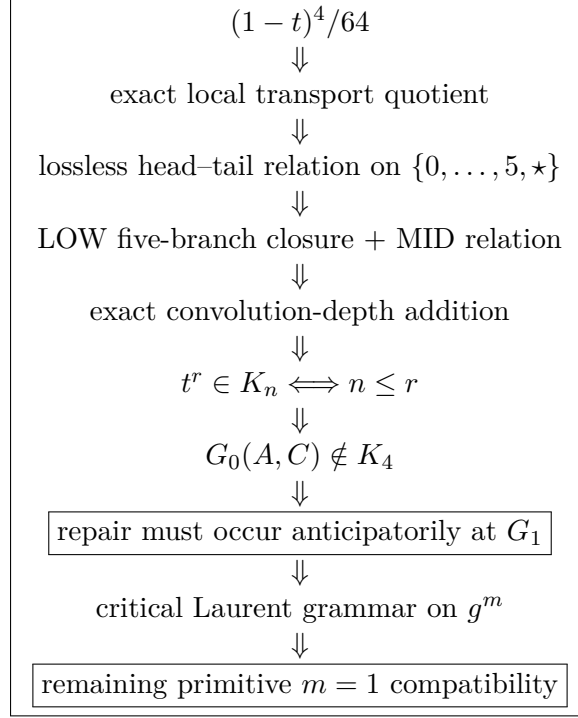
is not a viable zero-repair state.

Therefore all nonprimitive indices have been algebraically compressed, but the primitive boundary $m = 1$ remains load-bearing.

This is precisely the point where a positive anticipatory phase must replace the formal transition $g \rightarrow 1$.

11 Revised proof architecture

The exact status after the present reductions is



The previous target

“close G_0 recursively”

is therefore eliminated.

The revised target is:

Find one positive primitive modification of the $m = 1$ Laurent relation which replaces the forbidden child g^0 by a state in the same depth-labelled convolutional module, while preserving the finite head–tail compatibility relation.

Equivalently, the next theorem should be a

Pre-Boundary / Anticipatory Phase Lemma.

Once such a primitive block is obtained, common convolution propagates it through all $m > 1$ scales, and the anti-diagonal section converts the static positive decompositions into causal ones.

12 Ledger

Statement	Status
Correct phase rank 5 at $m = 2, 6$ for $m \geq 3$	PROVED
Exact $\{0, \dots, 5, \star\}$ head-tail quotient	PROVED
$\mathcal{L} = \mathcal{C}_{43/64}$ five-branch closure into CORE/ t	EXACT FINITE
MID relational escape $(B_1)_1 = (16c - 1)/12$	EXACT FINITE
Convolution-depth addition $K_m K_n \subseteq K_{m+n}$	PROVED
Terminal fuel $t^r \in K_n \iff n \leq r$	PROVED
Boundary atom estimate $a_n \leq 1/4 + (3/4)^{n+1}$	PROVED
Raw $G_0(A, C) \notin K_4$	PROVED
Exact interior deficit grammar	PROVED
Finite Laurent grammar for the critical g^m module	PROVED
Primitive $m = 1$ anticipatory repair	OPEN
Finite- f_k realization of the primitive Laurent repair	OPEN
Full causal addition lemma	OPEN
$O(\log n)$ upper bound	OPEN

Conclusion

The frontier has moved.

The problem is no longer to discover a large recursive boundary witness, nor to enlarge the six-dimensional phase vector, nor to repair a raw G_0 state deeper.

There are now two exact semantic contractions:

$$\boxed{\text{transport support} \longrightarrow \{0, \dots, 5, \star\},}$$

and

$$\boxed{\text{critical scale family} \longrightarrow \text{finite Laurent relation on } m.}$$

The first removes irrelevant spatial tail information. The second removes the infinite scale index.

Both reductions point to the same remaining object: one primitive, positive, anticipatory relation at the $m = 1$ boundary.

That is the next exact target.